UK SPACE AGENCY
SKILLS FOR SPACE INTERNSHIPS

MAGDRIVE

MMAT

The Magdrive Mission Analysis Tool

Diego Paniagua Silva BEng Aeronautics and Astronautics, University of Southampton

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Magdrive can now answer a customer manœuvre question in-house, in minutes, with a solver verified against NASA's GMAT.

THE PROJECT

Magdrive builds electric propulsion for small satellites. In this highly competitive market it is worthwhile for the company to be able to quickly and reliably assess the real life capabilities of their thrusters. This where mission analysis software like NASA's GMAT comes in. However, this kind of software is made to be used by someone with some mission analysis experience, and tends to have a GUI that is complex to understand.

The case for this project was therefore to equip Magdrive with a customisable but above all easy to use software to analyse customer missions, thruster specs, and potential bids.

WHY IN-HOUSE

- GMAT needs mission-analysis experience. MMAT does not.
- We own the code, so it can be adapted to Magdrive hardware.
- Engineering, client-facing teams and bid selection work from the same numbers.

IN, AND OUT

IN

Customer mission profile — start orbit, target orbit, constraints. Magdrive specifications — thruster, battery, solar arrays, masses.

OUT

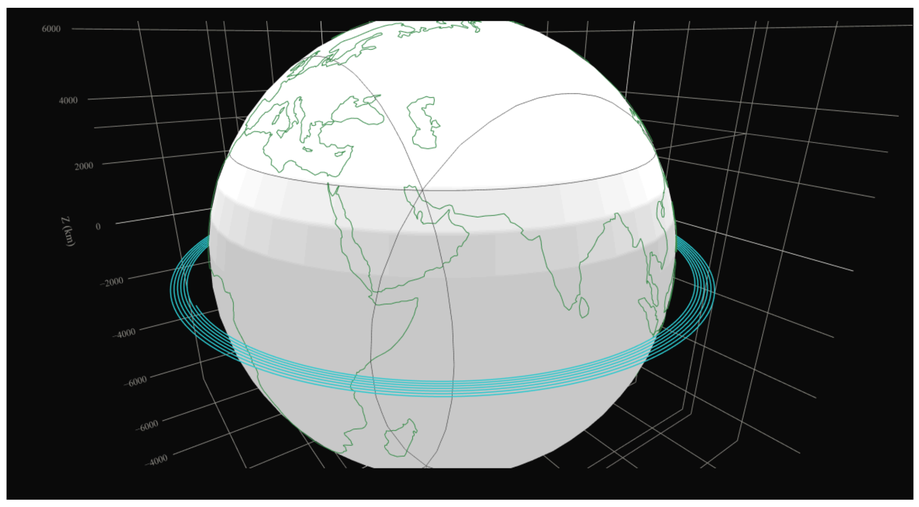
- Go / no-go on a bid or client opportunity
- Time of flight, ΔV, impulse, propellant, energy
- Trajectory, power and battery plots
- Comparison against competitors

WHAT I ACHIEVED

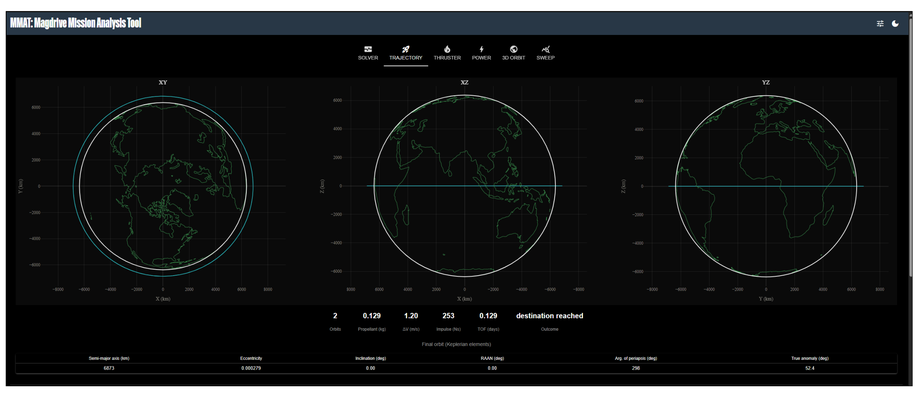
	EARLY JULY	TODAY
Entry point	Run a script, edit the file path by hand	**Desktop icon, native window, fields in a menu**
Plots	Static matplotlib windows	**Interactive, six tabs, some live during the run**
Test coverage	None	**Three suites, up to 225 scenarios**
Verification	30 GMAT scenarios, ~50% agreement	**225 GMAT scenarios, ~95% field agreement**
Mission structure	One manœuvre	**Chainable multi-phase mission plans**
Batch analysis	One run at a time	**Parallel sweep across a parameter range**
Distribution	Clone the repository	**One install command, then update with one more**



Live solver metrics update while the mission runs.



A low-thrust spiral orbit raise in three dimensions.



Trajectory in three planes, with final Keplerian elements.



Battery charge, power budget and eclipse irradiance.

A CUSTOMER QUESTION, ANSWERED

A ten-satellite constellation, 500 to 505 km circular, inclination 97.5°, three orbit raises across its lifetime. Can the Magdrive do it, and at what cost?

0.090	1.67	4.41	886	25	GO
Propellant, one raise (kg)	Time of flight (days)	ΔV (m/s)	Total impulse (Ns)	Battery cycles used	0.27 kg per spacecraft, three raises

WHAT I WOULD BUILD NEXT

WEEKS 1–2

Drag model

Unlocks long-duration low Earth orbit and deorbit scenarios.

WEEKS 3–9

More manœuvre types

Plane changes, station keeping, rendezvous and proximity operations.

WEEKS 9–16

Interplanetary transfers

Earth to Mars in one plan, on the same verified solver.

SKILLS I DEVELOPED

Space software engineering SCF-17, SCF-28
A native application over the solver, 27 modules, three test suites, packaged as an installer.

Verification and validation SCF-62, SCF-57
Designed and automated a 225-scenario GMAT campaign: test matrix, tolerances, ~95% agreement.

Flight mechanics SCF-1
Low-thrust orbit raising, J2, eclipse thruster behaviour, multi-phase plans.

Requirements and documentation SCF-43, SCF-8, SCF-29
Captured requirements from the engineer using the tool, wrote the user manual, worked under version control.

WHAT'S NEXT

Final year at Southampton, then at least one more internship or spell of industry work. After that a master's specialising in spacecraft engineering and flight dynamics, and a role in newspace.

THANK YOU

Tate Collins, my supervisor. Adam Baker, for the opportunity. Nick Clarke, for the software help. All the other Magdrive interns, for becoming great friends I hope to see again and maybe work with. Everyone else at Magdrive, for being an amazing team to work alongside.

LET'S CONNECT

Scan for my LinkedIn, I could not make it to the showcase but I'd love to hear what you are working on.

linkedin.com/in/dfps16